Week 4 – Performance & Load Testing

Performance Test Plan

Application Under Test: SauceDemo
Testing Tool: Apache JMeter

1. Introduction

Performance testing is conducted to evaluate the responsiveness, stability, and scalability of a web application under different levels of user load. This testing helps identify performance bottlenecks and ensures that the system performs efficiently when accessed by multiple users simultaneously.

In this project, performance testing was conducted on the SauceDemo web application using Apache JMeter to simulate multiple virtual users accessing the system.

2. Objectives

The objectives of this performance testing activity are:

- To measure the response time of the application
- To evaluate system behavior under simulated user load
- To identify potential performance issues
- To analyze system stability during concurrent user access
- To measure throughput and error rates

3. Scope of Testing

The scope of testing includes evaluating the performance of the **SauceDemo login page and homepage** under simulated user load.

The test focuses on the following activities:

- Accessing the homepage
- Simulating multiple users
- Monitoring response time and throughput

4. Test Environment

Component	Details
Application	SauceDemo
Testing Tool	Apache JMeter
Operating System	Windows 11
Processor	Intel Core i5
RAM	4 GB
Browser	Google Chrome
Protocol	HTTPS

5. Test Configuration

The following configuration was used for the load test.

Parameter	Value
Virtual Users	50
Ramp Up Period	10 seconds
Loop Count	5
Request Type	HTTP Request
Method	GET

6. Test Scenario

The test scenario simulates multiple users accessing the SauceDemo application simultaneously.

Steps executed during testing:

1. Launch Apache JMeter
2. Create a Test Plan
3. Add a Thread Group
4. Configure number of virtual users
5. Add HTTP Request sampler
6. Add listeners to monitor results
7. Execute the test

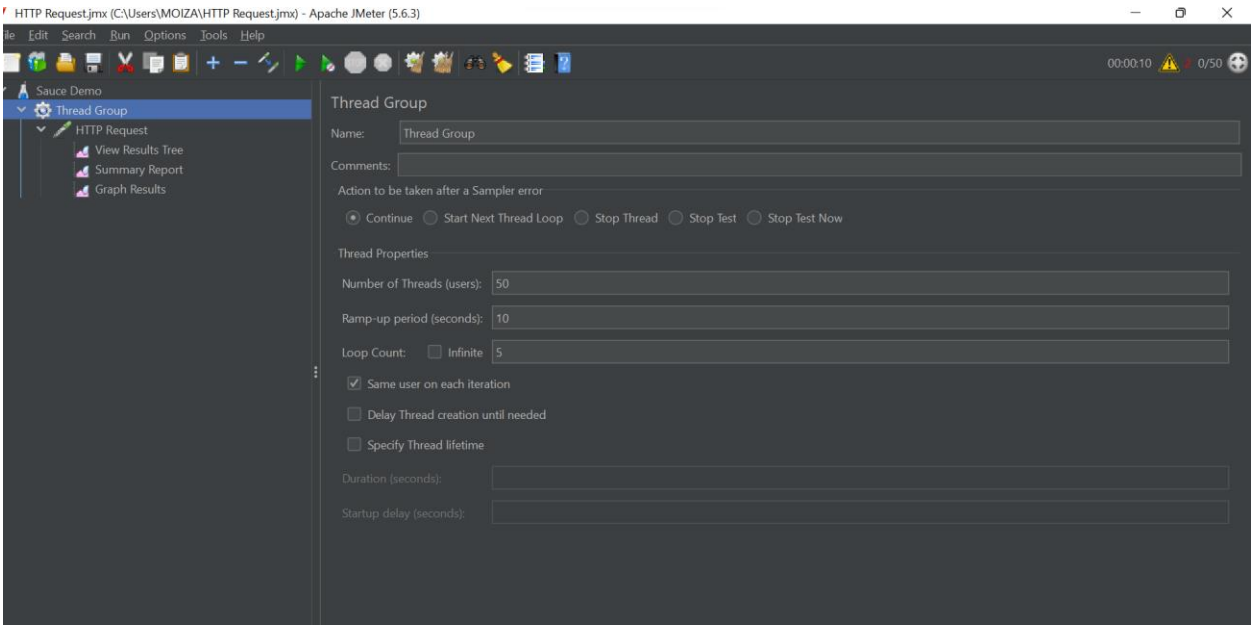

Figure 1:Test Plan and Thread Group

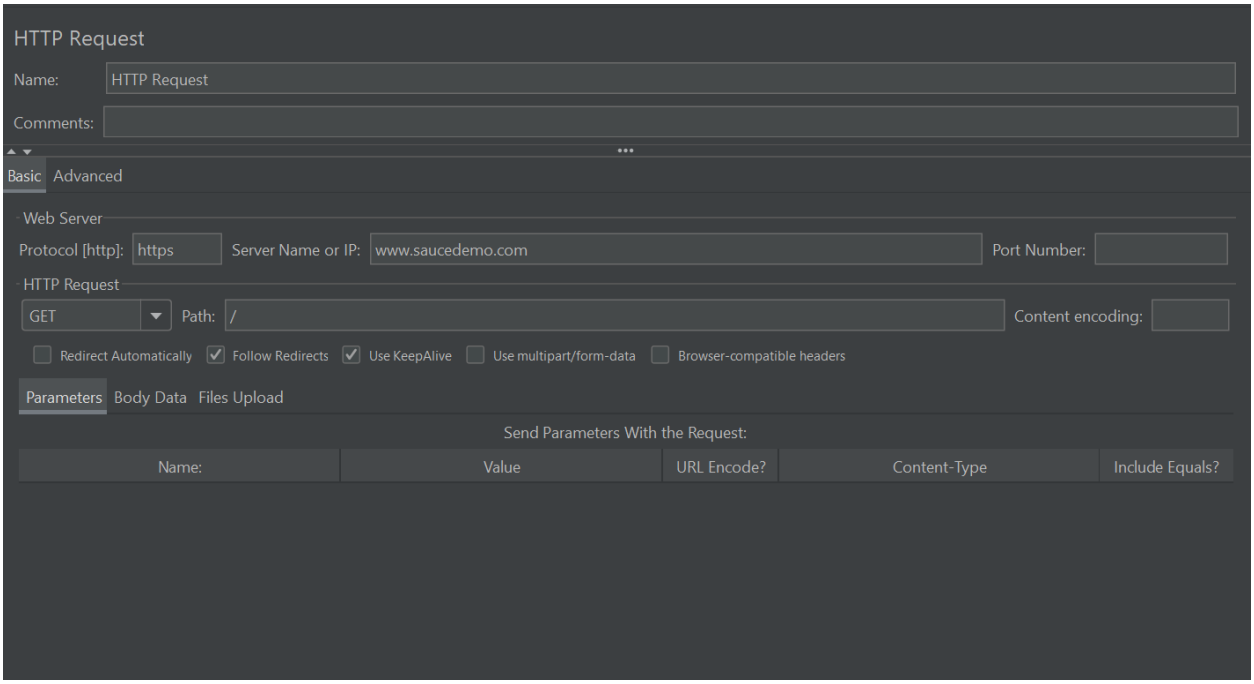

Figure 2:HTTPS Request